

The Self-Prioritization Effect Database with Standardized Meta-Data for Experimental Task

Cai Zhenxin¹, Tuo Liu, XXXX, Jie Sui^x, Hu Chuan-Peng^{1*}

¹ School of Psychology, Nanjing Normal University, 210024 Nanjing, China

*Email: hcp4715@hotmail.com

Abstract

The self-matching task, also known as self-tagging task, self-associative learning task, is widely used to investigate how self-related information is processed. In a typical self-matching task, participants first associate social labels (e.g., self, friend, and stranger) with geometric shapes (e.g., circle, triangle, and square) and then finish a matching task where they are asked to decide whether the shape-label pairings match the learned associations. Previous studies using this task have found faster and more accurate responses to self- than other-related pairings, i.e., the self-prioritization effect (SPE). Following the credibility revolution, a substantial number of these studies openly shared their data, created great potential for data re-using. However, there lacks of a standardized framework for sharing trial-level data generated by self-matching task, making it difficult to integrate existing open datasets. To fill the gap, we developed a three-level framework for organizing the metadata and trial-level data. We created machine-readable JSON files to document task implementations, including Physical Environment, Experimental Design, Block Structure, Trial Structure, and Stimulus Properties. We applied this framework to data available from 44 studies (70 experiments, 3,603 participants, and over 1.55 million trials). We demonstrated the value of the standardized open database through three examples: the effect of baseline conditions on SPE magnitude; the variability of SPE under nonmatch conditions; and the relationships between task details (duration of presenting time and numbere of trials) and SPE magnitude. Findings observed in all three examples could only be reliably detected using large aggregated datasets and can only be done with standardizing the meta-data across studies. Our standardized SPE database provides a valuable resource for the field. Moreover, our standardized framework for self-matching task is also valuable for developing similar databases for other cognitive tasks.

Keywords: Self-matching task, self-prioritization effect, open data, experimental design.

Introduction

The self plays a central role in cognition. Self-relevant information is processed faster and more accurately across perception, attention, and memory domains, which is known as self-prioritization effect (SPE) (Sui et al., 2012; Schäfer, Frings, et al., 2016; Constable et al., 2019). The self-matching task, also known as self-tagging task and self-associative learning task, has emerged as an important paradigm for investigating SPE. In a typical self-matching task, participant first associate geometric shapes with self- and non-self-relevant labels (e.g., "I", "Friend", and "Stranger") and then judge a shape-label pair presented on the screen match the learned association or not (Sui et al., 2012). Studies employed this task found better performance for self- than non-self-related stimuli (Liu et al., 2025; Schäfer, Wesslein, et al., 2016; Sui et al., 2012). Importantly, this effect has been observed across diverse cultural contexts (Constable et al., 2019; Grossmann & Jowhari, 2018) and different sensory modality (e.g., Schäfer, Wesslein, et al., 2016). This robust effect suggested a fundamental role of self-relevance in human cognition (Sui & Gu, 2017; Sui & Humphreys, 2015; Sui & Rotshtein, 2019).

In the era of credibility revolution, open data becomes a new norm (Houtkoop et al., 2018; Milham et al., 2018; Munafò et al., 2017). Specialized databases that aggregate data from multiple studies on particular phenomena provide unique opportunities for theory testing and discovery (Haaf et al., 2024; Rodriguez & Williams, 2022). Such resources allow researchers to examine effect sizes across diverse experimental conditions, identify moderating variables, and generate new hypotheses based on patterns that emerge only when data are aggregated across many studies (Cheung & Jak, 2016; Moreau & Gamble, 2022). The Attentional Control

Data Collection (ACDC) is a good example, which provides a structured open database for attentional control experiments that has facilitated meta-analyses and methodological advancements (Haaf et al., 2024). Similarly, the Confidence Database aggregates data from hundreds of experiments on confidence judgments and meta-cognitive performance into a unified repository (Rahnev et al., 2020). The Confidence Database enabled researchers to examine individual differences, compare computational models, and investigate theoretical questions that cannot be addressed using isolated datasets (Rahnev, 2025). While many SPE studies made their trial-level data publicly available, no standardized, large-scale dataset exists.

For cognitive paradigms such as the self-matching task, the lack of standardized meta-data and data organization slows down the reuse of open data scattered on the internet. For example, the primary outcome, reaction times, may be named differently across studies, such as RT, Response-Time, or Latency. Similarly, accuracy of each trial may use inconsistent coding schemes, such as 0 vs 1, True vs False, or Correct vs Error. The variable names of critical independent variables, such as the identities used in the self-matching task, may also be labelled differently across studies. More importantly, critical methodological information for implementing the task, such as the assignment of shapes to identities, stimulus presentation duration, trial structure, and participant exclusion criteria, is often absent from the raw data files and can only be found by reading the original papers. These issues substantially increase the effort required for reusing open data.

To address these issues, we developed a standardized framework for data organization and documentation, applied this framework to available open data, and formed the SPE database. To capture the methodological details systematically, we decomposed the self-matching task

into five components and documented information of each with standardized metadata files. We then collected all available datasets and manually standardized both behavioral variables and methodological descriptors to ensure cross-study comparability. In total, data from 44 studies, with 70 experiments, were included. The SPE database enables analyses that would be impossible for small sample size studies.

Method

Folder Structure

Following the FAIR principle (Wilkinson et al., 2016), we adopted a three-level folder structure for the database: root folder, study folder, and experiment folder (if there are multiple experiments). Each level is accompanied by standardized metadata encoded in machine-readable JSON files, facilitating both human interpretation and computational accessibility.

The root folder contains information spanning all studies included in the database. The study sub-folder contains information related to a specific study, defined as one or more experiments reported within a single publication, preprint, or thesis. The experiment folders include individual experimental data, typically representing a group of participants completing a specific version of the self-matching task across one or multiple sessions and assessment time points.

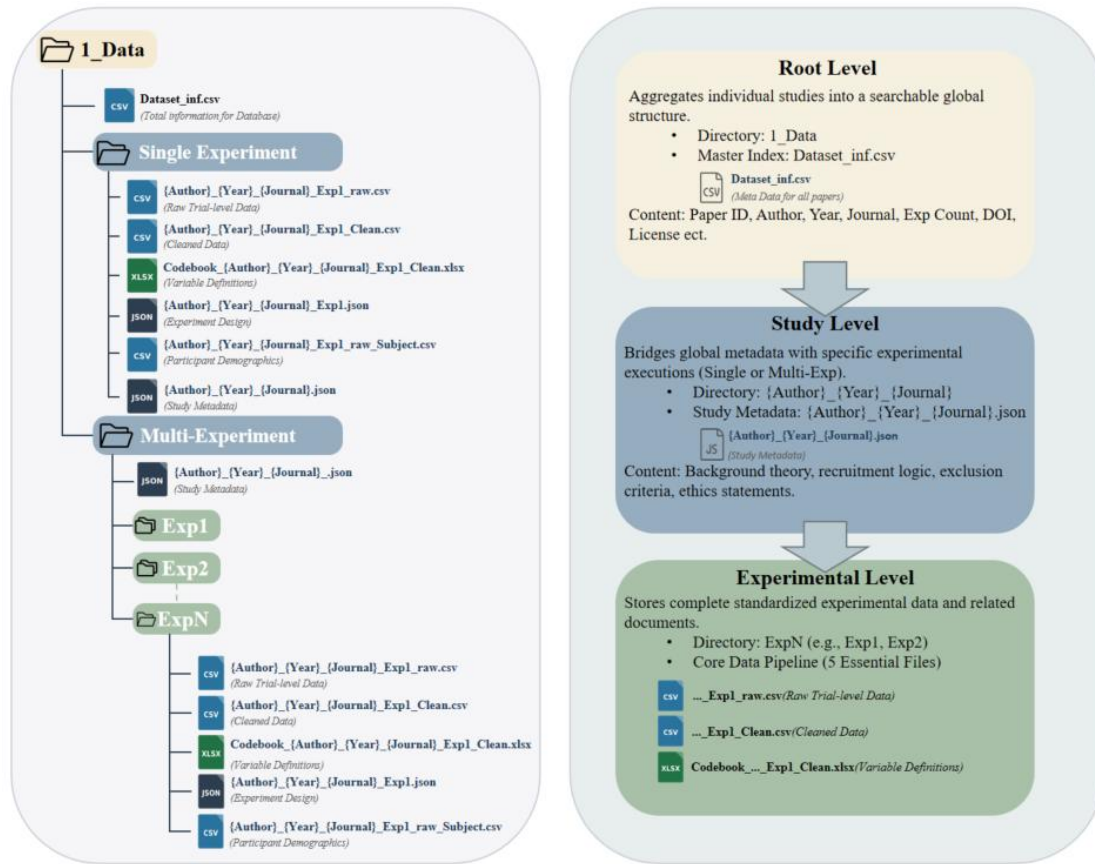


Figure 1. Database Folder Structure.

The root folder also contains a master metadata file ("Dataset_inf.csv"), which serves as the primary index for the entire database, and study-specific sub-folders (e.g., "Amodeo_2024_CABN"). Each row in this file represents one study and is uniquely identified by a 'Paper_Id', which is the primary key across different folders of the database. Each column of "Dataset_inf.csv" corresponds to a metadata field of the studies, such as authors, publication year, journal name (where applicable), Digital Object Identifiers (DOIs). Importantly, the study-specific folder ('Folder_Name') was included in "Dataset_inf.csv" for clear mapping between studies and their data folders.

The study-specific folder contains a JSON file, e.g., 'Amodeo_2024_CABN.json', and experiment-specific sub-folders. The JSON file contains study-level information such as theoretical background, participant recruitment strategies, participant recruitment procedures,

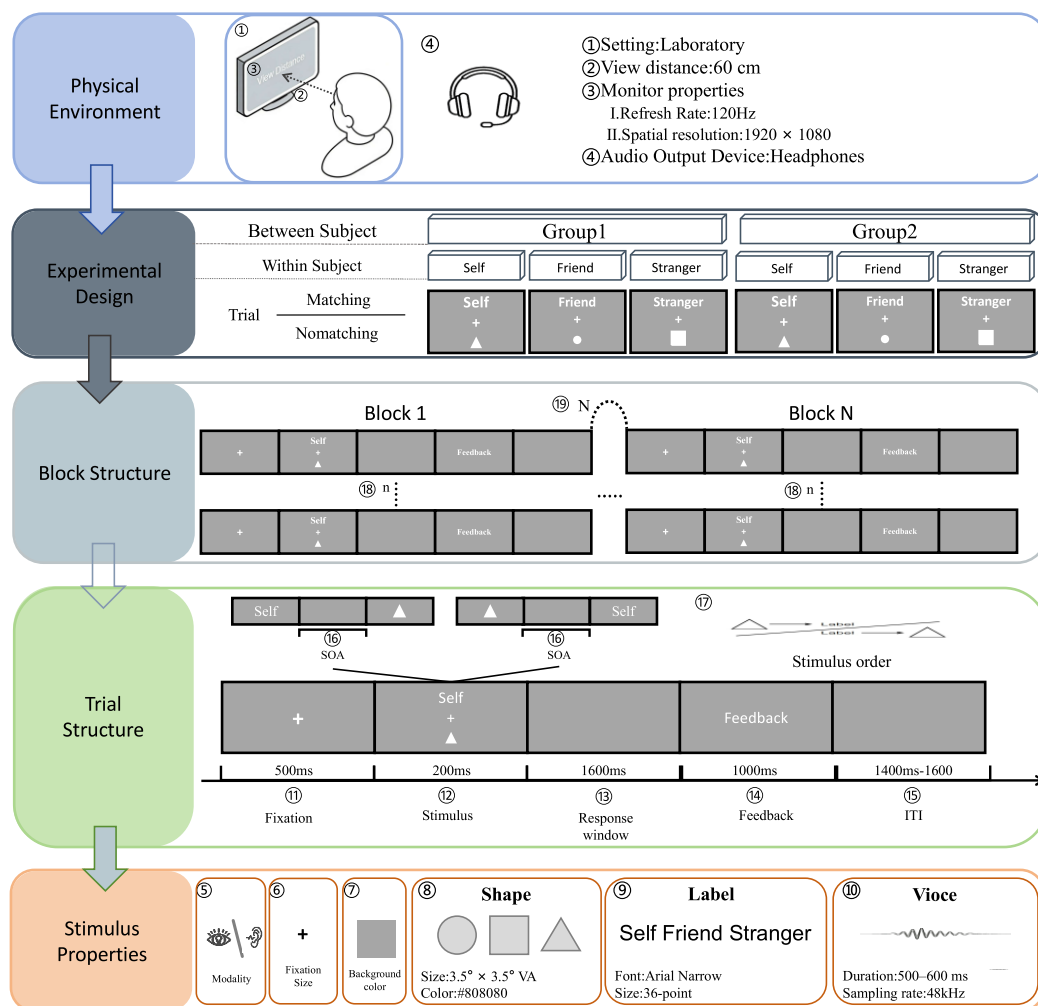
as well as inclusion and exclusion criteria. The experiment-specific subfolders correspond to the individual experiments reported within the the study. However, if a study contains only a single experiment, the experimental-level data are stored in the study-level folder directly and no experiment-specific subfolder was created.

For those studies reported more than one experiment, each experiment has its own subfolders (e.g., 'Exp1', 'Exp2'). Each experimental-level sub-folder include the following files: a CSV file for participants' demographic information (e.g., "Amodeo_2024_CABN_Exp1_raw_Subject.csv"), a JSON file for the experiment design (e.g., "Amodeo_2024_CABN_Exp1.json"), a CSV file for raw trial-level behavioral data file (e.g., "Amodeo_2024_CABN_Exp1_raw.csv"), a CSV file for the minimally preprocessed dataset (i.e., clean data, e.g., "Amodeo_2024_CABN_Exp1_Clean.csv"), and an xlsx file serves as the codebook for the clean data (e.g., "Amodeo_2024_CABN_Exp1_Clean.xlsx"). To facilitate efficient data retrieval and readability, all file names include four pieces of information — last name of the first author, publication year, journal abbreviation, and experiment identifier. Additionally, an optional fifth element specifies the content or function of the file (e.g., "_Subject" means information related to participants, "_raw" means unprocessed data from original studies). Together, these rules enhance data interoperability and facilitate large-scale secondary analyses.

Standardization of the task implementation

To further facilitate cross-study integration of datasets, we developed a framework for standardizing the implementation of the self-matching task. This framework dissects the task into five components: Physical Environment, Experimental Design, Block Structure, Trial

Structure, and Stimulus Properties (Figure 2). These five components zoom from the environment in which the experiment was done to the details of stimuli. Also, these details may also play a role in cognitive processes (e.g., Gold & Shadlen, 2007; Ratchliff et al., 2016), and allow nuanced cross-study analyses. We standardized all these five components of each experiment and stored them in the experiment-specific json file (e.g., "Amodeo_2024_CABN_Exp1.json") whenever relevant information is available.



```

Data > Amedeo_2024_CABN > {} Amedeo_2024_CABN_Exp1.json > ...
1  {
2    "exp1": {
3      "schema_version": "2",
4      "Collected_date": "/",
5      "Physical_Environment": {
6        "Location": "Ghent University (implied by ethical committee approval)",
7        "Setting": "Electrically shielded chamber (for EEG recording)",
8        "Equipment": {
9          "Presenting": "/",
10         "Monitor": "/",
11         "Software": "E-prime Studio, version 2.0.10.248"
12       },
13       "Viewing_distance": "/"
14     },
15     "Experimental_Design": {
16       "Conditions": "/"
17     },
18     "Block_Structure": {
19       "Practice_trials": "24 trials (repeatable up to 3 times if accuracy <70%)",
20       "Block_number": "6 blocks",
21       "Trial_number": "360 trials total (60 trials per block)"
22     },
23     "Trial_Structure": {
24       "Stimulus_order": "Simultaneously",
25       "SOA": "0 ms (simultaneous presentation)",
26       "Fixation_duration": "500 ms",
27       "Stimulus_duration": "Practice: 1000 ms; Test: 200 ms",
28       "Response_deadline": "Practice: 2400 ms; Test: 1600 ms",
29       "ITI": "Variable 1400-1600 ms",
30       "Feedback_duration": "1000 ms (practice phase only)"
31     },
32     "Stimulus_Properties": {
33       "Modality": "Visual",
34       "Fixation": "/",
35       "Shape": "3.7° × 3.7°",
36       "Label": "36-point Arial Narrow",
37       "Colors": {
38         "Stimulus": "White",
39         "Background": "Light gray"
40       }
41     },
42     "detail": ""
43   }
44 }

```

Figure 2 (a) Standardized implementation framework and five-component decomposition of the self-matching task; (b) an example of the experiment-specific json file.

The Physical Environment variables describe the overall environmental conditions under which participants completed the task, including basic settings (e.g., laboratory-based vs. online administration), conditions relevant to multimodal stimulus presentation. For visual stimuli, these encompass viewing distance, monitor refresh rate, display resolution, brightness (Lin et al., 2023). For auditory stimuli, these include hardware specifications (e.g., headphones vs. speakers), sound pressure levels, and ambient noise (Kirk & Cunningham, 2025). These

parameters determine how sensory information (e.g., visual or auditory signals) is physically presented to participants and they may modulate the sensory input of information processing (Bridges et al., 2020; Gold & Shadlen, 2007).

The Experimental Design variables define the structure of the study, including both between-subject factors (e.g., cultural group) and within-subject manipulations (e.g., Identity and Matching conditions). What distinguishes this level from the other levels is that design variables specify *what* is being manipulated and compared for hypothesis testing, rather than *how* the task is procedurally implemented, although these aspects may overlap. This distinction is consistent with established reporting guidelines that differentiate methodological details from inferential variables (Appelbaum et al., 2018).

The Block Structure describe the block structure of the task, including number of blocks as well and details within a block: trial sequence and composition. Block Structure captures how multiple trials are organized within a block and often may strongly influence the strategies adopted by participants (Bogacz et al., 2006) or result in serial dependency effect (Cicchini et al., 2024). Usually, trials in self-matching task are pseudo-randomized such that no more than three identical trial types are presented consecutively.

The Trial Structure zoom in each trial and describe how each trial is implemented, including fixation duration, stimulus exposure, stimulus onset asynchrony (SOA), response window, and inter-trial interval (ITI). These parameters govern the temporal dynamics of stimuli presentation and are closely related to cognitive mechanisms such as evidence accumulation and response preparation (Ratcliff et al., 2016).

Finally, the Stimulus Properties describe details of the stimuli, including their modality,

their visual/auditory properties, and attributes of labels. The defining boundary of this level is whether a piece of information specifies *what* information is presented rather than *how* it is physically delivered, with the latter belong to Trial Structure.

Summary of data included in SPE database

The Self-Matching Database is hosted on the Open Science Framework (OSF) website (XXX) and Science Data Bank (XXX). At the time of publication, the database contained 48 studies with 70 datasets, aggregating 3,603 participants across 1,554,083 trials (See Table 1 for details).

Table 1*Overview of Studies Included in the SPE Database*

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
1	Amodeo_2024_CABN	L Amodeo et al. (2024)	Exp1	Belgium	Dutch	37 (///)	geometric shape	360	NA	Lab Experiment
2	Amodeo_2024_CABN	L Amodeo et al. (2024)	Exp1	Belgium	Dutch	30 (///)	geometric shape	360	NA	Lab Experiment
3	Atzeni_2026_PsychRes	Atzeni (2026)	Exp1	Italy	Italian	140	geometric shape	200 (per session; duplicated/extra runs: 3 T2 + 4 T3 participants)	NA	Online Experiment
4	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp1	Austria	English	132 (0/132)	geometric shape	180	CC BY 4.0	Lab Experiment
5	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp2	Austria	German	35 (0/35)	geometric shape	360	NA	Lab Experiment
6	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp2	Austria	German	35 (0/35)	geometric shape	360	NA	Lab Experiment
7	Bukowski_2021_ActaPsych	Bukowski et al. (2021)	Exp2	Austria	German	33 (0/33)	geometric shape	360	NA	Lab Experiment
8	Constable_2019_JEPHPP	Constable et al. (2019)	Exp1	Hungary	Hungarian	28 (12/16)	geometric shape	288	No License	NA
9	Constable_2019_JEPHPP	Constable et al. (2019)	Exp2	Hungary	Hungarian	28 (6/22)	geometric shape	288	No License	NA
10	Constable_2019_JEPHPP	Constable et al. (2019)	Exp3	Hungary	Hungarian	28 (8/14)	geometric shape	288	No License	NA

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
11	Constable_2019_JEPHPP	Constable et al. (2019)	Exp4	Hungary	English	40 (14/26)	geometric shape	640	No License	NA
12	Constable_2020_ActaPsych	Constable & Knoblich (2020)	Exp1	Hungary	Hungarian	48 (///)	geometric shape	240	No License	Lab Experiment
13	Constable_2020_ActaPsych	Constable & Knoblich (2020)	Exp1	Hungary	Hungarian	44 (///)	geometric shape	240	No License	Lab Experiment
14	Constable_2021_CogEmo	Constable et al. (2021)	Exp1	United Kingdom	English	28 (16/12)	grey scale squares	384	No License	Lab Experiment
15	Constable_2021_CogEmo	Constable et al. (2021)	Exp1	United Kingdom	English	28 (9/19)	grey scale squares	384	No License	Lab Experiment
16	Constable_2021_CogEmo	Constable et al. (2021)	Exp2	Hungary	English	28 (13/15)	grey scale squares	192	No License	Lab Experiment
17	Constable_2021_CogEmo	Constable et al. (2021)	Exp2	Hungary	English	28 (15/13)	grey scale squares	192	No License	Lab Experiment
18	Dalmaso_2024_ConsciousCog	Mario Dalmaso et al. (2024)	Exp1	Japan	Japanese	40 (19/21)	face	360	CC BY 4.0	Lab Experiment
19	Dalmaso_2024_ConsciousCog	Mario Dalmaso et al. (2024)	Exp2	Italy	Italian	40 (16/24)	face	360	CC BY 4.0	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
20	Feldborg_2021_IJERPH	M Feldborg et al. (2021)	Exp1	UK	English	48 (5/43)	geometri c shape and face	320	CC BY 4.0	Online Experiment
21	Feldborg_2021_IJERPH	M Feldborg et al. (2021)	Exp1	UK	English	54 (12/40)	geometri c shape and face	320	CC BY 4.0	Online Experiment
22	Golubickis_2021_ActaPsych	Golubickis & Macrae (2021)	Exp1	United Kingdo m	English	30 (9/21)	geometri c shape	360	NA	Online Experiment
23	Golubickis_2021_ActaPsych	Golubickis & Macrae (2021)	Exp2	United Kingdo m	English	25 (8/17)	geometri c shape	360	NA	Online Experiment
24	Haciahmet_2023_Psychophysiol	Haciahmet et al. (2023)	Exp1	German y	German	40 (8/32)	geometri c shape	240	CC BY 4.0	Lab Experiment
25	Hobbs_2023_PsychMed	Hobbs et al. (2023)	Exp1	United Kingdo m	English	144 (31/112)	geometri c shape and face and currency	360	CC BY 4.0	Lab Experiment
26	Hu_2020_CollabraPsy	Hu et al. (2020)	Exp1	China	Chinese	44 (19/27)	geometri c shape	480	No License	Lab Experiment
27	Hu_2023_psyarxiv	Hu et al. (2023)	Exp1	China	Chinese	23	geometri c shape	90	On request	Lab Experiment
28	Kirk_2025_BritJPsy	Neil W. Kirk et al. (2025)	Exp1	United Kingdo m	English	35 (35/0)	voice	216	CC BY 4.0	Online Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
29	Kirk_2025_BritJPsy	Neil W. Kirk et al. (2025)	Exp2	United Kingdom	English	126 (90/0)	voice	216	CC BY 4.0	Online Experiment
30	Kolvoort_2020_HumBrainMap	IR Kolvoort et al. (2020)	Exp1	Canada	English	31 (17/14)	geometric shape	100	NA	Lab Experiment
31	Lee_2023_Cognition	Lee et al. (2023)	Exp1	United Kingdom	English	57	geometric shape	96	CC BY 4.0	Online Experiment
32	Lee_2023_Cognition	Lee et al. (2023)	Exp2	United Kingdom	English	65	geometric shape	96	CC BY 4.0	Online Experiment
33	Liang_2022_HumBrainMap	Liang et al. (2022)	Exp1	China	English	35 (13/22)	geometric shape	360	CC BY-NC-ND	Lab Experiment
34	Liang_2022_HumBrainMap	Liang et al. (2022)	Exp1	China	English	38 (22/16)	geometric shape	360	CC BY-NC-ND	Lab Experiment
35	Liang_2022_HumBrainMap	Liang et al. (2022)	Exp1	China	English	36 (13/23)	geometric shape	360	CC BY-NC-ND	Lab Experiment
36	Liu_2023_CogRes	Liu et al. (2023)	Exp1	UK	English	302 (132/161)	face	64	NA	Online Experiment
37	Martinez-Perez_2024_ConsciousCog	V Martínez-Pérez et al. (2024)	Exp2	Spain	Spanish	32 (5/27)	geometric shape	240	CC BY 4.0	Lab Experiment
38	Mcivor_2021_EJN	L McIvor et al. (2021)	Exp1	United Kingdom	English	20 (6/14)	geometric shape and face	960	CC BY 4.0	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
39	Mcivor_2021_EJN	L McIvor et al. (2021)	Exp1	United Kingdom	English	20 (2/18)	geometric shape and face	960	CC BY 4.0	Lab Experiment
40	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp1	Israel	Hebrew	13 (3/10)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
41	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp2	Israel	Hebrew	27 (9/18)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
42	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp3	Israel	Hebrew	28 (12/15)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
43	Navon_2021_psyarxiv	Navon & Makovski (2021)	Exp4	Israel	Hebrew	27 (9/18)	geometric shape	360	CC0 1.0 Universal	Lab Experiment
44	Orellana-Corrales_2020_ExpPsych	G Orellana-Corrales et al. (2020)	Exp1	Germany	German	38 (9/29)	geometric shape	128	NA	Lab Experiment
45	Orellana-Corrales_2020_ExpPsych	G Orellana-Corrales et al. (2020)	Exp2	Germany	German	32 (2/3)	geometric shape	128	NA	Lab Experiment
46	Orellana-Corrales_2020_ExpPsych	G Orellana-Corrales et al. (2020)	Exp3	Germany	German	36 (7/29)	geometric shape	128	NA	Lab Experiment
47	Orellana-Corrales_2021_APP	G Orellana-Corrales et al. (2021)	Exp1	Germany	German	34 (9/25)	geometric shape	128	No License	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
48	Orellana-Corrales_2021_APP	G Orellana-Corrales et al. (2021)	Exp2	Germany	German	34 (11/23)	letter string	128	No License	Lab Experiment
49	Orellana-Corrales_2023_QJEP	G Orellana-Corrales et al. (2023)	Exp1	Germany	German	71 (46/25)	geometric shape	140	NA	Online Experiment
50	Orellana-Corrales_2023_QJEP	G Orellana-Corrales et al. (2023)	Exp1	Germany	German	65 (42/23)	geometric shape	140	NA	Online Experiment
51	Pan_2025_unpub	Pan (2025)	Exp1	China	Chinese	40	geometric shape	NA	/	NA
52	Perrykkad_2022_BMCPsych	Perrykkad et al. (2022)	Exp1	United States	English	334	geometric shape	360	CC BY 4.0	Online Experiment
53	Qi_2025_SciData	Qi (2025)	Exp1	China	Chinese	134 (57/77)	geometric shape	360	CC BY 4.0	Lab Experiment
54	Qian_2020_QJEP	Qian et al. (2020)	Exp1	China	Chinese	26 (10/16)	geometric shape	576	No License	Lab Experiment
55	Qian_2020_QJEP	Qian et al. (2020)	Exp2	China	Chinese	26 (7/19)	geometric shape	200	No License	Lab Experiment
56	Schaefer_2019_JCogPsych	Schaefer & Frings (2019)	Exp1	Germany	German	32	geometric shape	144	CC-BY-SA 4.0	Lab Experiment
57	Schaefer_2019_JCogPsych	Schaefer & Frings (2019)	Exp2	Germany	German	36	geometric shape	144	CC-BY-SA 4.0	Lab Experiment
58	Schaefer_2019_JCogPsych	Schaefer & Frings (2019)	Exp3	Germany	German	35	geometric shape	144	CC-BY-SA 4.0	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
59	Smith_2024_Cortex	Smith & Ferguson (2024)	Exp1	UK	English	48 (11/5)	face	NA	CC BY-NC-ND	Lab Experiment
60	Sui_2014_APP	Sui et al. (2014)	Exp1	China	Chinese	24	geometric shape	360	/	Lab Experiment
61	Sui_2014_APP	Sui et al. (2014)	Exp2	China	Chinese	18	geometric shape	360	/	Lab Experiment
62	Sui_2014_APP	Sui et al. (2014)	Exp3	China	Chinese	22	geometric shape	360	/	Lab Experiment
63	Sui_2014_APP	Sui et al. (2014)	Exp4	China	Chinese	20	geometric shape	360	/	Lab Experiment
64	Sui_2014_unpub	Sui (2014)	Exp1	China	Chinese	24	geometric shape	40	/	NA
65	Sui_2015_unpub	Sui (2015)	Exp1	China	Chinese	20	geometric shape	480	NA	NA
66	Sui_2015_unpub	Sui (2015)	Exp2	China	Chinese	20	geometric shape and face	480	NA	NA
67	Sui_2023_ConsciousCog	J Sui et al. (2023)	Exp1	UK	English	20 (12/8)	geometric shape	960	CC BY-NC-ND	Lab Experiment
68	Sun_2026_DataExp	Sun et al (2026)	Exp1	China	Chinese	—	geometric shape	NA	No License	NA
69	Svensson_2022_PsychRes	Svensson et al. (2022)	Exp1	United Kingdom	English	20 (6/14)	geometric shape	200	NA	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
70	Svensson_2022_PsychRes	Svensson et al. (2022)	Exp2	United Kingdom	English	24 (8/16)	geometric shape	400	NA	Lab Experiment
71	Svensson_2022_PsychRes	Svensson et al. (2022)	Exp3	United Kingdom	English	25 (7/18)	geometric shape	400	NA	Lab Experiment
72	Svensson_2023_QJEP	Svensson et al. (2023)	Exp1	UK	English	65 (20/45)	geometric shape	120	No License	Online Experiment
73	Vicovaro_2022_JEPHPP	Vicovaro et al. (2022)	Exp1	Italy	Italian	30 (14/16)	random dots	480	CC BY 4.0	Mixed (Lab + Online)
74	Vicovaro_2022_JEPHPP	Vicovaro et al. (2022)	Exp2	Italy	Italian	52 (11/36)	random dots	240	CC BY 4.0	Mixed (Lab + Online)
75	Vicovaro_2022_JEPHPP	Vicovaro et al. (2022)	Exp2	Italy	Italian	52 (5/42)	random dots	240	CC BY 4.0	Mixed (Lab + Online)
76	Vicovaro_2024_PeerJ	Vicovaro et al. (2024)	Exp1	Italy	English	64 (26/38)	geometric shape	240	NA	Lab Experiment
77	Vicovaro_2024_PeerJ	Vicovaro et al. (2024)	Exp2	Italy	English	64 (16/48)	geometric shape	240	NA	Lab Experiment
78	Wang_2016_JEPHPP	Wang et al. (2016)	Exp1	China	English	21 (3/18)	geometric shape	648	CC BY 3.0	NA
79	Wang_2016_JEPHPP	Wang et al. (2016)	Exp2	China	English	25 (10/15)	geometric shape	648	CC BY 3.0	NA
80	Wozniak_2018_PLOS	Wozniak et al. (2018)	Exp1	Hungary	English	19 (9/9)	face	672	CC BY 4.0	Lab Experiment
81	Wozniak_2018_PLOS	Wozniak et al. (2018)	Exp2	Hungary	English	20 (9/9)	face	672	CC BY 4.0	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
82	Wozniak_2020_PLOS	Wozniak et al. (2020)	Exp1	Australia	English	24 (12/12)	face	270	NA	Lab Experiment
83	Wozniak_2020_PLOS	Wozniak et al. (2020)	Exp2	Australia	English	24 (12/12)	face	270	NA	Lab Experiment
84	Wozniak_2020_PLOS	Wozniak et al. (2020)	Exp3	Australia	English	24 (12/12)	face	270	NA	Lab Experiment
85	Wozniak_2022_PsychRes	Wozniak et al. (2022)	Exp1	Hungary	English	23 (7/11)	geometric shape	360	CC BY 4.0	Lab Experiment
86	Wozniak_2022_PsychRes	Wozniak et al. (2022)	Exp2	Hungary	English	21 (6/12)	geometric shape	360	CC BY 4.0	Lab Experiment
87	Wozniak_2022_PsychRes	Wozniak et al. (2022)	Exp3	Hungary	English	18 (0/18)	geometric shape	360	CC BY 4.0	Lab Experiment
88	Xu_2022_CurrPsych	Xu et al. (2022)	Exp1	China	Chinese	55 (27/28)	geometric shape	360	On request	Lab Experiment
89	Xu_2022_CurrPsych	Xu et al. (2022)	Exp1	China	Chinese	50 (24/26)	geometric shape	360	On request	Lab Experiment
90	Zhang_2023_NeuroImage	Yongfa Zhang et al. (2023)	Exp1	China	Chinese	346 (172/174)	geometric shape	96	CC BY-NC-ND	Lab Experiment
91	Zhang_2024_PsychJ	Zhang et al. (2024)	Exp1	China	English	43 (///)	geometric shape	216	NA	Lab Experiment
92	Zhang_2024_PsychJ	Zhang et al. (2024)	Exp2	China	English	36 (///)	geometric shape	120	NA	Lab Experiment
93	Zhang_2026_JNeurosci	Zhang (2026)	Exp1	China	English	59 (27/32)	geometric shape	450 (5 blocks x 90; incomplete blocks in 17 subjects)	NA	Lab Experiment

Num	ID	Study	Exp	Country	Language	N (M/F)	Stimulus	Trials	License	Exp_Implement
94	Zhang_2026_JNeurosci	Zhang (2026)	Exp1	China	English	32 (14/18)	geometric shape	450 (5 blocks x 90; incomplete blocks in 17 subjects)	NA	Lab Experiment

Note. k = number of experiments. The diversity of countries, languages, stimulus modalities, and experimental designs ensures that future re-analyses can address a broad range of scientific questions—from the robustness and generalisability of the SPE across cultures and stimulus formats, to the identification of experimental design parameters that moderate the magnitude of the effect.

As we mentioned in the section *Folder Structure*, data files include both raw trial-level data files shared from each team (e.g., "Amodeo_2024_CABN_Exp1_raw.csv") and minimally preprocessed data files (e.g., "Amodeo_2024_CABN_Exp1_Clean.csv"), both are stored in experiment-specific subfolders (see Section XXX). The corresponding codebook (e.g., "Amodeo_2024_CABN_Exp1_Clean.xlsx") describes each column (see Figure 3). Basic information such as participant index (Subject), matching condition (Matching; matching or nonmatching), shape stimulus (Shape), label stimulus (Label), reaction time in milliseconds (RT_ms) and seconds (RT_sec), and response accuracy (ACC) are included in all experiments. The identity columns for both shape and label in data files were standardized into five categories: Self, Close, Acquaintance, Celebrity, Stranger, NonPerson (Sun et al., 2023). For clarity, we kept the original identity names ("Origin_Identity"), the English-translated identity names ("English_Identity"), and the standardised identity categories ("Standardised_Identity"). The minimally preprocessed data files may also include additional fields that describe the experimental design, such as block number, trial number within block, experimental phase or session, specific stimulus features (e.g., face gender, voice accent, emotional valence), and participant response keys. We expect these minimally preprocessed data files will be valuable for researchers interested in re-using this database.

Variable_name	Variable_description	Variable_value	Variable_category
Subject	Participant number	Number	Numerical
Block	Block number in the experiment; each block may contain practice or experimental trials	Number	Numerical
Trial	Trial number within the block	Number	Numerical
Shape	Visual shape stimulus presented in the trial	T. bmp;S. bmp;C. bmp	Categorical
Label	Text label presented for matching with the shape	Vreemde;Ikzelf;Bekende	Categorical
Matching	Type of matching task for the trial; indicates whether the shape and label match or not	Matching;Nonmatching	Categorical
Label_Origin_Identity	Original identity associated with the label	Vreemde;Ikzelf;Bekende	Categorical
Label_English_Identity	English translation of the label's identity	Stranger;Self;Friend	Categorical
Label_Standardized_Identity	Standardized identity for the label used across experiments	Stranger;Self;Friend	Categorical
Shape_Origin_Identity	Original identity associated with the shape stimulus	Stranger;Self;Friend	Categorical
Shape_English_Identity	English translation of the shape's identity	Stranger;Self;Friend	Categorical
Shape_Standardized_Identity	Standardized identity for the shape used across experiments	Stranger;Self;Close	Categorical
Response	Participant's response in the trial	n;m	Categorical
RT_ms	Reaction time for the response, measured in milliseconds	Number	Numerical
RT_sec	Reaction time for the response, measured in seconds	Number	Numerical
ACC	Accuracy of the participant's response	0;1	Categorical

Figure 3 Raw codebook of experimental dataset variables

Example uses of the SPE Database

The Self-Prioritization Database provides a valuable resource for both methodological and theoretical investigations of the self-prioritization effect. For example, the standardized schema allows researcher to develop and test cognitive models of self-prioritization effects; investigate how experimental design parameters shape these effects; clarify the cross-cultural universality and cultural specificity of self-prioritization; and evaluate the robustness and generalizability of findings across laboratory-based and online settings. To demonstrate the unique value of large-scale standardized datasets, we present three examples addressing questions that cannot be adequately examined using data from individual studies alone.

The Magnitude of Self-Prioritization Effect Varies Across Social Distance

The self-prioritization effect (SPE) is typically quantified as the performance difference between self-referential and other-referential conditions. However, different categories of others have been used as baseline conditions when estimating SPE magnitude. Existing studies have yielded inconsistent estimates of the self-prioritization effect (SPE) magnitude. Nevertheless, a systematic examination of how baseline selection affects SPE magnitude remains lacking. Leveraging the large-scale sample size of the SPE database, we compared the effect size of the SPE across five standardized baselines (Sun et al., 2023): Non-person, Stranger, Celebrity, Acquaintance, and Close Other. Standardized effect sizes (Cohen's d) were computed at the participant level by contrasting self-related trials against each baseline condition separately, using a pooled standard deviation. A linear mixed-effects model without intercept was then fitted to the participant-level Cohen's d values, with participants as a random intercept and data source as an additional variance component. From this model, 10,000

multivariate normal samples were drawn from the estimated fixed-effects distribution to obtain the posterior-like distributions of each identity's SPE magnitude. For consistency, we use positive Cohen's d for SPE (i.e., faster reaction times or higher accuracy for self relative to other identities).

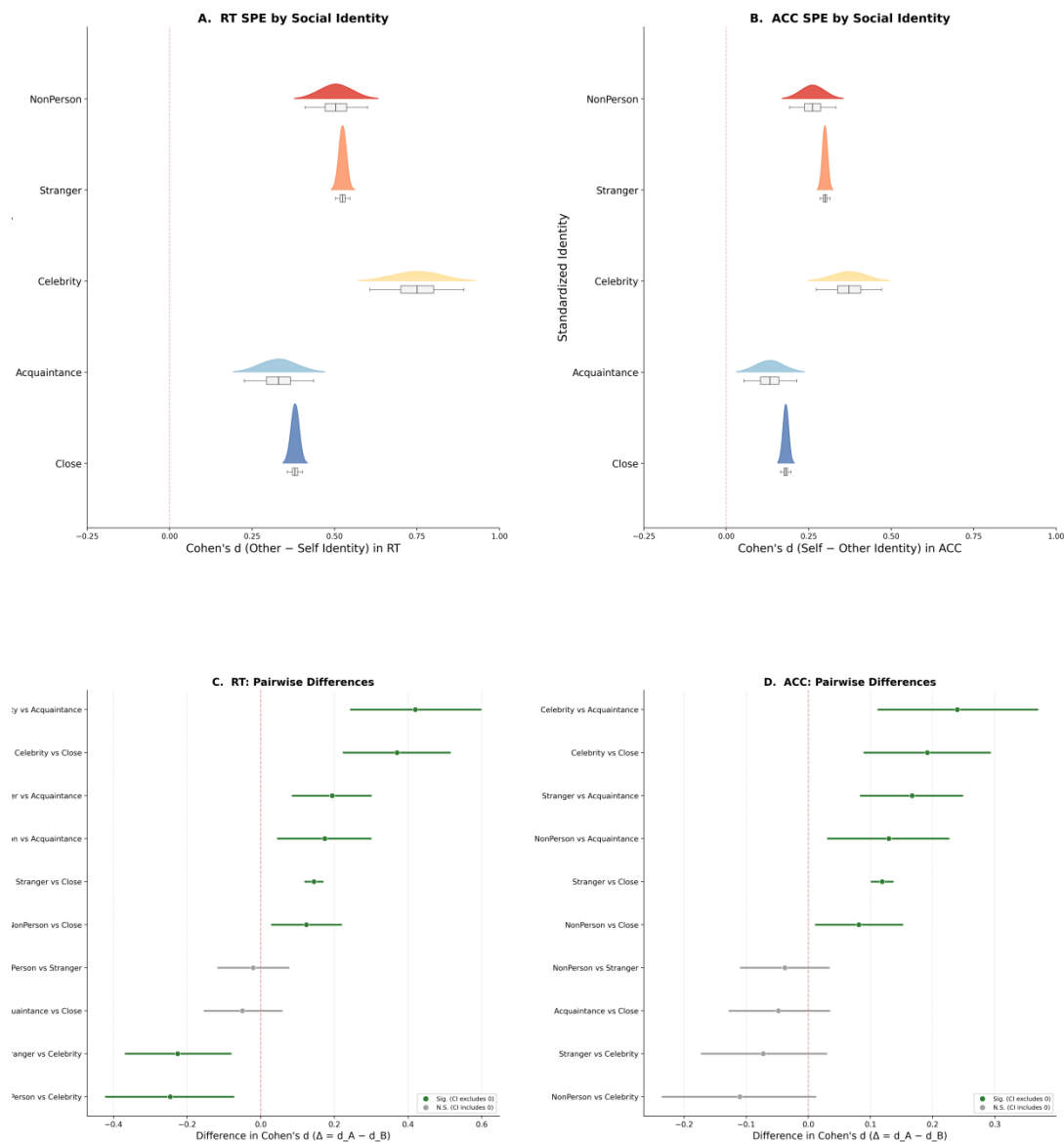


Figure 4 Self-prioritization effects along the social distance continuum and pairwise comparisons across social identities..

For reaction time (RT), robust SPEs were observed across all five baseline conditions, with all effect sizes significantly greater than zero (all 95% CIs excluded zero; all $ps < .001$).

The magnitude of the SPE differed systematically across baselines (see Figure 3). To test whether the magnitude statistically differ between SPE calculated by different baselines, we computed these effect sizes pairwise ($\Delta = d_A - d_B$) using the model-based sampling approach. We infer the difference exist if the bootstrap 95% CI does not include zero (see Figure 3C and 3D).

For RT, 8 of the 10 pairwise comparisons were statistically significant. The Celebrity-based SPE was significantly larger than that those derived from Acquaintance ($\Delta = 0.42$, 95% CI [0.24, 0.60], $p < .001$), Close others ($\Delta = 0.37$, 95% CI [0.22, 0.51], $p < .001$), Non-person ($\Delta = 0.25$, 95% CI [0.07, 0.42], $p = .006$), and Strangers ($\Delta = 0.23$, 95% CI [0.08, 0.37], $p = .001$). The Stranger-based SPE was significantly larger than that for Acquaintances- ($\Delta = 0.19$, 95% CI [0.09, 0.30], $p < .001$) and Close-other-based SPE ($\Delta = 0.14$, 95% CI [0.12, 0.17], $p < .001$). Similarly, the Non-person-based SPE is greater than Acquaintance- ($\Delta = 0.17$, 95% CI [0.05, 0.30], $p = .007$) and Close-other-based SPE ($\Delta = 0.12$, 95% CI [0.03, 0.22], $p = .009$). However, Non-person- versus Stranger-based SPE ($\Delta = -0.02$, 95% CI [-0.12, 0.08], $p = .678$) as well as Acquaintance- versus Close-other-based SPE ($\Delta = -0.05$, 95% CI [-0.15, 0.06], $p = .348$) did not reach statistical significance.

A similar pattern emerged for accuracy (ACC). Six of the 10 pairwise comparisons reached statistical significance. Celebrity-based SPEs were significantly larger than those derived from Acquaintance ($\Delta = 0.24$, 95% CI [0.11, 0.37], $p < .001$) and Close-other-based SPE ($\Delta = 0.19$, 95% CI [0.09, 0.29], $p < .001$); Stranger-based SPE is larger than Acquaintance- ($\Delta = 0.17$, 95% CI [0.08, 0.25], $p < .001$) and Close-other-based SPE ($\Delta = 0.12$, 95% CI [0.10, 0.14], $p < .001$); Non-person-based SPE is larger than Acquaintance- (Δ

= 0.13, 95% CI [0.03, 0.23], $p = .009$) and Close-other-based SPE ($\Delta = 0.08$, 95% CI [0.01, 0.15], $p = .023$). The remaining four pairs are not significant: Celebrity- versus Stranger ($\Delta = 0.07$, 95% CI [-0.03, 0.17], $p = .164$), Celebrity versus Non-person ($\Delta = 0.11$, 95% CI [-0.01, 0.23], $p = .079$), Non-person versus Stranger ($\Delta = -0.04$, 95% CI [-0.11, 0.03], $p = .298$), and Acquaintance versus Close other ($\Delta = -0.05$, 95% CI [-0.13, 0.03], $p = .243$).

Taken together, these findings indicate that the estimated magnitude of the SPE varies systematically as a function of the baseline condition employed. Across both RT and ACC, three tiers of SPE magnitude emerged: Celebrity stimuli elicited the strongest SPE, followed by Strangers and Non-person stimuli (which did not differ from each other), and finally Close others and Acquaintances (which also did not differ from each other). This pattern is largely consistent with the self-culture theory (Rose, 1991) and inclusion of other in the self-concept (Aron et al., 1992) in social psychology, suggesting that SPE is closely linked to the distance to the self.

Presence of Self-Prioritization Effect Under Nonmatching Conditions

Previous studies have reported inconsistent findings regarding the presence of self-prioritization effects (SPEs) under nonmatching conditions. Although several studies have observed reliable nonmatching SPEs (Kirk & Cunningham, 2025), others have failed to detect such effects (Desebrock & Spence, 2021; Golubickis & Macrae, 2021). One potential explanation for these inconsistencies is that nonmatching SPEs are substantially smaller than matching SPEs and therefore require larger samples to be detected reliably. An alternative is that the SPEs under nonmatching conditions were contaminated by the involvement of self-reference in the baseline condition that used for calculating SPE.

Leveraging the large sample size of the current database (72 experiments, 853,015 nonmatch trials, 1,676 participants), we estimated the SPE using a bootstrap resampling approach (500 iterations per sample size, with replacement) that allows evaluation of effect size stability across increasing sample sizes. Recall that in the self-matching task, each trial presents a shape–label pair. Thus, for matching trials, the shape and label are congruent (e.g., a shape previously associated with "Self" paired with the word "Self"), making the distinction between self-related and other-related trials straightforward. For **nonmatching conditions** (also referred to as *incongruent* or *mismatching* trials), however, the shape and label convey conflicting identity information (e.g., a shape for "Self" paired with the word "Stranger"), creating difficulty in defining what constitutes a "Self" trial.

To address this ambiguity, we adopted two alternative operationalizations of the self-prioritization effect (SPE) under nonmatching conditions: shape-based SPE or label-based SPE. **The shape-based SPE is the difference** between trials where the shape is "Self" and the label is non-Self and trials where the shape is non-self condition(s) and the label does not match the shape. **The label-based SPE is difference** between trials where the label is "Self" and the shape is non-self and trials where the label is non-self condition(s) and the shape does not match the label.

Critically, neither the shape-based nor the label-based operationalization guarantees a clean self versus non-self contrast, because nonmatching trials classified as "non-self" may still contain self-referential information. For example, in an experiment includes only two identities, Self and Stranger, the nonmatching trials are either "Self(shape)-Stranger(label)" or "Stranger(shape)-Self(label)". Under both conditions, part of the information is self-referential,

making it impossible to construct a pure contrast between self- versus non-self-referential conditions. A pure self versus non-self contrast becomes feasible only in experiments involving three or more identities. For example, under the shape-based operationalization, self-related trials can be defined as "Self(shape)-Other1(label)/Other2(label)" whereas non-self trials can be defined as "Other1(shape)-Other2(label)". In this case, the comparison trials contain no self-referential information, thereby providing a rigorous estimate of nonmatching SPEs.

To fully exploit the available datasets while preserving interpretability, we implemented both conservative and liberal approach for estimating the SPE for nonmatching trials. The **conservative approach** only include pure self versus non-self contrasts: only experiments involving at least three distinct identities were eligible for inclusion. At the trial level, additional filtering procedures were applied. For shape-based SPE, nonmatching trials in which the shape corresponded to a non-self identity but the label referred to the Self were excluded. Conversely, for label-based SPE, trials in which the label corresponded to a non-self identity but the shape was associated with the Self were removed. Using these criteria, 63 experiments were included in the conservative analyses. The final datasets comprised 303,331 trials for the shape-based SPE and 199,271 trials for the label-based SPE.

In contrast, the **liberal approach** maximized dataset coverage by retaining all eligible experiments and nonmatching trials regardless of the presence of residual self-referential information in the comparison condition. To ensure a fair comparison between shape-based and label-based SPE for nonmatching trials, we aligned sample-size by using the minimum N of the shape- and label-based SPE for both conservative and liberal approaches (see table 3 for the included sample size and number of trials for different approaches).

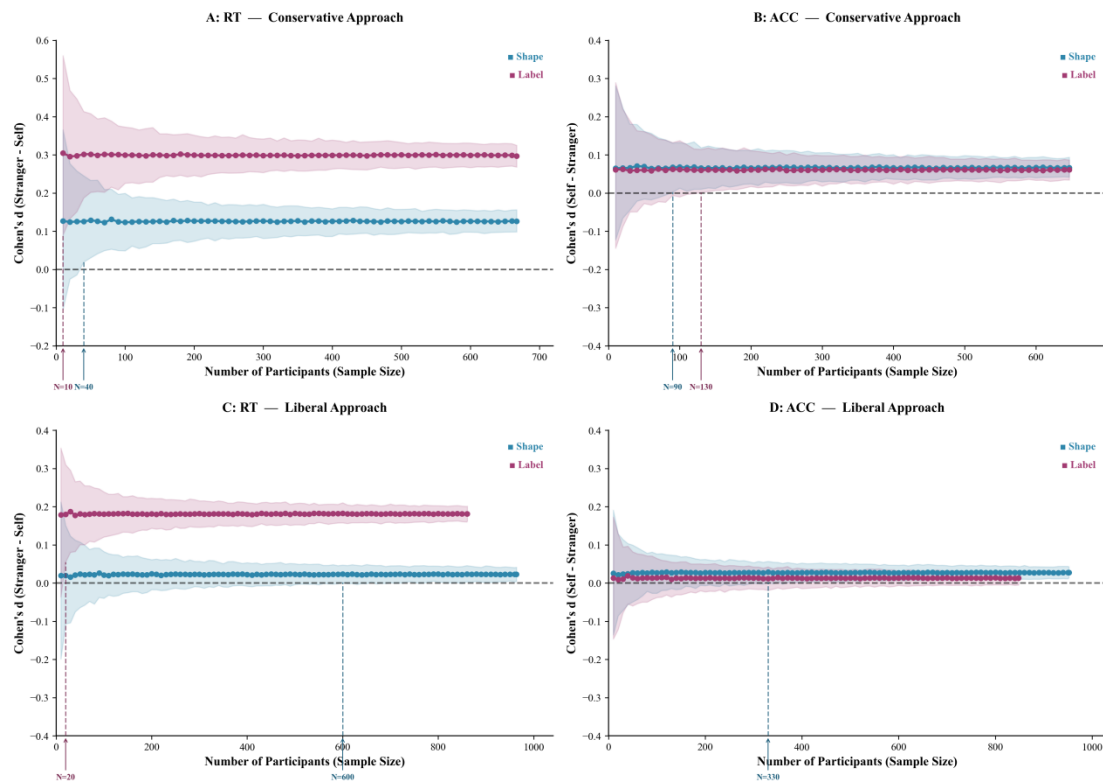


Figure 5 Bootstrap estimation of the self-prioritization effect under nonmatching conditions.

A and B, the conservative approach; C and D, the liberal approach.

Table 3

Summary of Liberal Approach Results

	Measure	Dimension	N _{Max}	Cohen's d	95% CI	N _{min}
Liberal Approach	RT	Shape	964	0.023	[0.004, 0.041]	630
	RT	Label	861	0.182	[0.163, 0.202]	20
	ACC	Shape	952	0.028	[0.012, 0.043]	280
	ACC	Label	848	0.013	[-0.004, 0.031]	inf
Conservative Approach	RT	Shape	667	0.126	[0.101, 0.152]	40
	RT	Label	667	0.298	[0.270, 0.324]	10
	ACC	Shape	647	0.066	[0.040, 0.092]	90
	ACC	Label	647	0.061	[0.036, 0.088]	120

Note. For RT, $d = (\text{Mean_Stranger} - \text{Mean_Self}) / \text{Pooled_SD}$, such that positive values indicate faster responses to self trials (SPE). For ACC, $d = (\text{Mean_Self} - \text{Mean_Stranger}) / \text{Pooled_SD}$, such that positive values indicate higher accuracy for self trials (SPE). N min is the smallest bootstrap sample size at which the 95% CI no longer includes zero, as marked by vertical dashed lines in Figure 4.

Using the conservative approach, robust SPE is observed for both reaction time (RT)

and accuracy (ACC). For RT, the label-based operationalization yielded a moderate SPE ($d = 0.298$, 95% CI [0.270, 0.324], $N = 667$), indicating faster responses for nonmatching trials in which the label corresponded to the Self relative to those in which the label referred to other identities. The corresponding 95% confidence interval first excluded zero at merely $N = 10$, indicating that the label-based RT SPE can be detected even in relatively small samples (see Figure 4A). The **shape-based** operationalization also revealed a significant RT SPE, although the effect was substantially smaller ($d = 0.126$, 95% CI [0.101, 0.152], $N = 667$). In this case, the 95% confidence interval first excluded zero at $N = 40$ (see Figure 4A). For accuracy (ACC), both label-based and shape-based operationalizations produced relatively small but statistically significant effects. The shape-based SPE was estimated at $d = 0.066$, 95% CI [0.040, 0.092], $N = 647$, with convergence at $N = 90$ (see Figure 4B). Similarly, The label-based SPE reached significance, $d = 0.061$, 95% CI [0.036, 0.088], $N = 647$, with convergence at $N = 120$ (see Figure 4B). Taking together, we found that SPE exist for both RT and accuracy for nonmatching trials when using a conservative approach. Also, **label-based SPE was larger than the shape-based SPE** for RT (difference in Cohen's $d \approx 0.172$, 95% CI [0.132, 0.214]) but not for ACC (differences in Cohen's $d \approx -0.006$, 95% CI [-0.043, 0.031]).

When applying the liberal approach, the label-based SPE was found for reaction time (RT), although the effect size was smaller than that observed under the conservative approach, $d = 0.182$, 95% CI [0.163, 0.202], $N = 861$. The corresponding 95% confidence interval first excluding zero at $N = 20$ (see Figure 4C), indicating that the label-based RT effect remained detectable with relatively modest sample sizes. In contrast, the **shape-based** SPE was markedly attenuated under the liberal approach. Although the overall effect reached statistical

significance ($d = 0.023$, 95% CI [0.004, 0.041], $N = 964$), the 95% confidence interval did not consistently exclude zero until $N = 630$ (see Figure 4C), suggesting that the effect was extremely small and difficult to detect reliably in typical experimental samples. For accuracy, the shape-based operationalization also yielded a small but statistically significant SPE ($d = 0.028$, 95% CI [0.012, 0.043], $N = 952$), with confidence intervals consistently excluding zero from $N = 280$ (see Figure 4D). However, the label-based ACC condition was not significant ($d = 0.013$, 95% CI [-0.004, 0.031], $N = 848$), as its confidence interval consistently included zero across the full range of sample sizes (see Figure 4D).

Collectively, leveraging the large sample size of the current database, we found that SPE exist for nonmatching trials when conservative operational criteria were applied. Moreover, SPE under nonmatching trials was larger for in reaction time than accuracy, a pattern that is consistent with previous findings in the SPE literature (Liu et al., 2025). These findings underscores the importance of both adequate statistical power and transparent operational definitions when investigating the SPE under nonmatching conditions.

The Association Between Presenting Duration / Trial Number and magnitude of SPE

The third example illustrates a unique advantage of the current database: the standardized coding of methodological variations in task implementation across studies. Specifically, we examined two implementation parameters that varied across experiments: **stimulus presentation duration** and **trial number**. Stimulus duration was chosen because shorter presentation times may create time pressure for participants and alter their processing strategies (Lemaire & Brun, 2016; Ratcliff et al., 2016). Despite its potential theoretical relevance, variation in stimulus duration has received little attention in the SPE literature and

is often treated as a minor procedural detail. Trial number was examined because the amount of behavioral data collected from each participant may influence the reliability and stability of SPE estimates, consistent with broader concerns regarding measurement precision in cognitive tasks (Liu et al., 2025). Both implementation parameters were treated as continuous variables. For each analysis, we estimated the association between the implementation parameter and SPE magnitude (Cohen's d_z) using bootstrap Spearman correlations with 10,000 resamples. We report the observed Spearman ρ , bootstrap 95% confidence intervals (2.5th–97.5th percentiles), and bootstrap p values (two-tailed; statistical significance assessed at $\alpha = .05$).

We found a robust negative association between stimulus presentation duration (treated as a continuous variable, range: 100 – 1,500 ms, $n = 40$ datasets) and SPE magnitude. Bootstrap Spearman correlation revealed a significant negative relationship for both RT ($\rho = -.493$, bootstrap 95% CI $[-.729, -.197]$, $p_{\text{boot}} = .001$) and ACC ($\rho = -.467$, bootstrap 95% CI $[-.682, -.174]$, $p_{\text{boot}} = .002$; Figures 5A and 5B). These results indicate that SPE magnitude decreased monotonically as stimulus duration increased. Specifically, shorter presentation times were associated with larger SPE effect sizes, whereas longer presentation times tended to yield progressively smaller effects. Notably, the bootstrap confidence intervals for both RT and ACC were entirely below zero, providing evidence for a reliable negative association between stimulus duration and the magnitude of self-prioritization.

We also observed significant positive associations between the maximum number of trials in a dataset and SPE magnitude (Figures 5C and 5D). For reaction time (RT), bootstrap Spearman correlation revealed a moderate positive relationship ($\rho = .374$, bootstrap 95% CI $[.098, .599]$, $p_{\text{boot}} = .010$). A comparable association was observed for accuracy (ACC) (ρ

= .389, bootstrap 95% CI [.079, .634], $p_{\text{boot}} = .016$). To determine whether these associations were driven by a small number of datasets with extreme trial counts, we conducted a sensitivity analysis excluding the three datasets with the largest trial numbers (see red dots in Figure 6C and 7D). The Spearman correlations remained significant for both RT ($\rho = .331$, $p = .032$) and ACC ($\rho = .361$, $p = .019$), confirming that the positive association was not attributable to outliers. These results consistently indicate that datasets with larger maximum trial numbers tended to yield larger SPE estimates, with the association slightly stronger for accuracy than for reaction time.

Together, these exploratory findings indicate that SPE magnitude varies systematically as a function of how the task is implemented. Specifically, shorter stimulus presentation duration are associated with larger SPE magnitudes, whereas greater trial numbers are associated with larger SPE magnitudes. These findings should be interpreted with caution. The relationships between SPE magnitude and stimulus presentation duration or number of trials may not be strictly linear and may depend on important boundary conditions. For example, the SPE may change when the stimulus presentation duration becomes subliminal. Similarly, the benefits associated with increasing trial numbers may plateau beyond a certain point as participants become exhausted. Future studies are needed to experimentally manipulate these implementation parameters to establish their causal contributions to SPE magnitude. Nevertheless, our findings highlight the importance of considering methodological details when studying self-prioritization. More broadly, they demonstrate the value of systematically coding task implementation characteristics in large-scale databases. Variables that are often treated as minor procedural details may meaningfully contribute to variability in observed SPE

estimates across studies. Careful documentation of these implementation features may therefore facilitate more precise theory testing, improve cross-study comparisons, and enhance the reproducibility of findings in the self-prioritization literature.

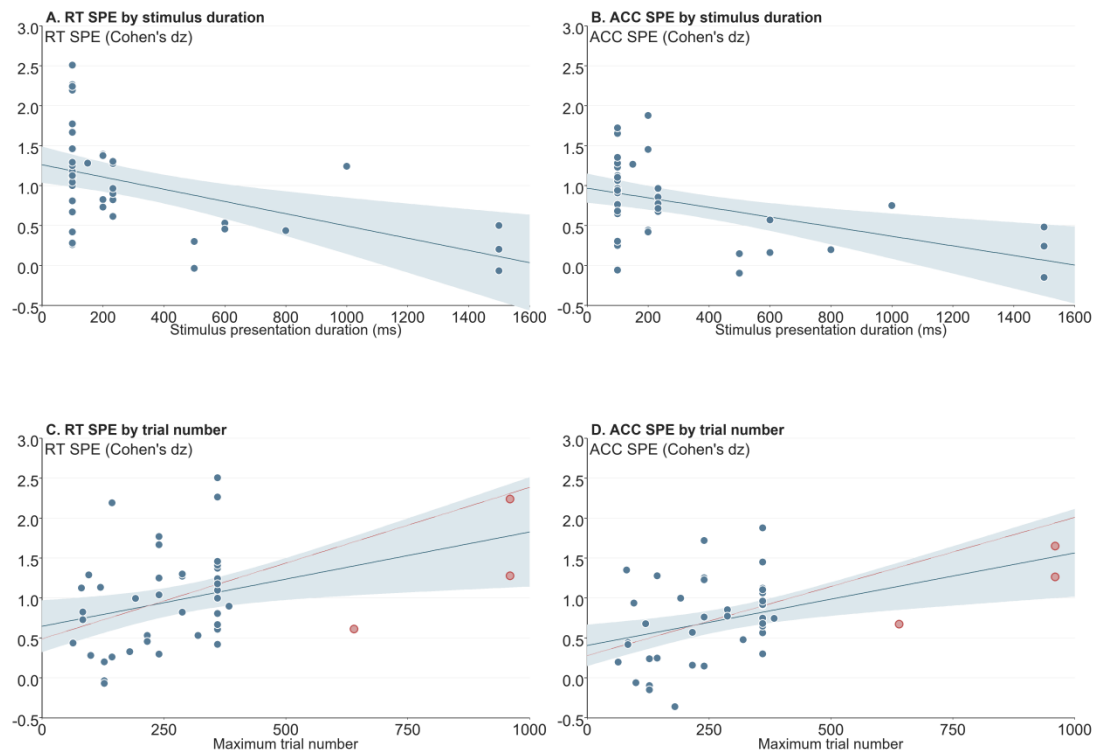


Figure 6 *Exploratory moderators of the self-prioritization effect under matching conditions.*

Discussion

The present study introduces both standardized data organization framework for self-matching task and the first manually curated, large-scale standardized database for the self-prioritization effect (SPE). Guided by the FAIR principles of data management (Wilkinson et al., 2016), we developed a three-level folder structure consisting of root-, study-, and experiment-level folders, each included standardized, machine-readable JSON metadata files.

This folder structure strikes a balance between human interpretability and machine readability, thereby facilitating data sharing, reuse, and large-scale secondary analyses. Moreover, we developed a standardized, five component framework for annotating the metadata of task implementation details of self-matching task, which will facilitate cumulative, reproducible, and data-intensive research on self-prioritization. By applying the folder structure and standardized framework for annotating task details to available data, we curated a large database with unified meta data, which allows researchers to address theoretical and methodological questions that would be difficult or impossible to examine using individual studies alone.

The SPE database provides a valuable resource for understanding of SPE. Through three illustrative examples, we demonstrated how standardized, large-scale datasets can address theoretical and methodological questions that can not be solved by single small sample study. We found that the magnitude of the SPE varies across different baselines, the existence of SPE under nonmatch conditions varied with operationalizations, and that task implementation details such stimulus duration and trial numbers correlated with the magnitude of SPE. Together, these findings exemplify the value of such a well-maintained large-scale database for the field.

It worth noting that the SPE database provides the first standardized framework for annotation of data generated from self-matching task. We decomposes the implementation of the self-matching task into five core components: Physical Environment, Experimental Design, Block Structure, Trial Structure, and Stimulus Properties. This decomposition follows explicit operational criteria grounded in classical information-processing theory (Gold & Shadlen,

2007) and formal decision-making models (Ratcliff et al., 2016), thereby providing clear boundaries for assigning variables to each component and avoiding arbitrary categorization. By including comprehensive details in metadata, our database substantially enriches the methodological information available to researchers in ways that are not typically provided in published articles, as demonstrated by our third example.

Importantly, this standardized framework can extend beyond the self-matching task. Many laboratory-based cognitive tasks share similar structural components, our framework can serve as a template for standardizing metadata regarding the implementation details in other tasks. Reporting these details in machine-readable format will increase the transparency of empirical studies and provide more nuanced data beyond the current best practices recommended by Journal Article Reporting Standards for quantitative research in psychology (Appelbaum et al., 2018). These rich information will be valuable for both peer-review process, post-publication review, and meta-research. The current cognitive tasks may also vulnerable to the jingle-jangle fallacy (e.g., Sun, Wang, Wen, & Chuan-Peng, 2023), reporting these details will help to understand how the details of implementing task will change the cognitive construct under the investigation.

Limitations of the current database should be acknowledged. First, the curation of SPE database relied heavily on manual extraction, limiting the scalability and efficiency of the standardized framework for task implementation. Future efforts may learn from existing standards such as the Brain Imaging Data Structure (BIDS, Poldrack et al., 2024) or Hierarchical Event Descriptors (HED, Robbins et al., 2022) to improve the level of. Second, the current version of SPE database focuses exclusively on behavioral measures. Future

versions should aim to include multimodal datasets, such as EEG (e.g., Sui et al., 2023), fMRI (e.g., Sui et al., 2013), using established standards for neuroimaging data organization. Third, the current SPE database only included data that are permitted by authors or the data license, potentially introducing bias into the database, such as the participants' representativeness (Ghai, 2024; Ghai et al., 2025; Liu et al., 2025). We made all code and script of the present database open available to facilitate future larger-scale collaborative projects that will expand the SPE database.

To conclude, we believe the SPE database will serve as a value resource for community-driven initiatives including large-scale collaborative projects and crowdsourced replication efforts. By combining standardized data structures with comprehensive methodological metadata, the database provides an infrastructure capable of supporting coordinated investigations of self-prioritization across diverse experimental settings and populations. More broadly, this work illustrates how standardized, FAIR-compliant databases can transform fragmented collections of open datasets into cumulative scientific resources that accelerate theory development, improve reproducibility, and promote collaborative discovery in cognitive psychology.

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